6 **(a)** Since reading decreases, magnetic force is upwards. B1

From Fleming's left hand rule, magnetic field points/directed into the page. B1

(b) magnetic force on coil $F = \frac{(125.76-101.35)\times10^{-3}}{2} \times 9.81 = 0.11973$ N C1

$F = NBIL$ M1

$B = \frac{0.11973}{60\times6.0\times0.085}$

$B = 3.91 \times 10^{-3}$ T A1

(c) forces on PS & RQ act horizontally / act in opposite directions / cancels off B1

no effect on the reading B1